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CS499

Artifact 2: Treasure Hunt

Algorithms and Data Structures

Description of the Artifact

The second artifact I selected is my Treasure Hunt project from CS370. This project involved reinforcement learning and Q-learning. The program uses an agent that learns how to make decisions based on rewards and penalties.

This project was different from many of my other programming assignments because the goal was not simply to tell the computer exactly what to do. Instead, the program was designed to learn from its actions and improve its decisions through training.

Why I Selected This Artifact

I selected the Treasure Hunt project because it shows my experience with algorithms and gives an example of me working with a more advanced computer science concept.

Reinforcement learning was challenging for me because it was a different way of thinking about programming. I had to understand concepts such as exploration and exploitation and how the Q-learning process uses information from previous decisions.

For my enhancement, I added additional tracking to the training process. The enhanced version records information such as the number of steps taken during each episode, exploration and exploitation activity, epsilon information, and average steps. I also created a training history so that the results could be reviewed instead of only looking at the final training result.

What I Learned From the Enhancement

The biggest thing I learned from this enhancement was that having more information about an algorithm makes it easier to understand what the algorithm is doing.

Before the enhancement, it was harder to see what was happening during each episode of training. By tracking the number of steps and other training information, I had a better way to see how the training was progressing.

I also learned more about the difference between exploration and exploitation. The agent needs to try different choices while also using choices that it has already learned are useful. Finding a balance between those two things is an important part of reinforcement learning.

The enhancement also made me think more about how algorithms can be evaluated. It is not enough to say that an algorithm ran successfully. It is helpful to collect information that can show how it performed and whether changes to the training process made a difference.

Challenges and Feedback

One of the biggest challenges with this project was understanding reinforcement learning and getting the training code to work correctly. I also had issues with the training environment while working through the project.

Working through those problems helped me become better at troubleshooting. Instead of immediately assuming the whole program was wrong, I learned to look at the error and figure out which part of the program was causing the problem.

The enhancement also required me to think about what information would actually be useful to track. I did not want to add information just for the sake of adding it. The goal was to make the training process easier to understand.

Course Outcomes

This artifact directly supports the course outcome related to designing and evaluating computing solutions using algorithmic principles. It demonstrates my experience working with an algorithm and then improving the way its performance could be observed.

It also demonstrates my ability to use computer science tools and techniques to solve a problem. The enhancement gave me additional experience analyzing the behavior of an algorithm instead of only focusing on whether the program produced an output.